

Domperidone: From Molecule to Market to Monitoring

An Integrated Regulatory, Pharmacovigilance & Data Analysis Case Study

Self-Directed Portfolio Project

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B.Pharm | Executive Diploma in Drug Regulatory Affairs (in progress)

About This Project

This is an independent, self-directed training project, undertaken to apply and demonstrate skills developed through formal coursework: an Executive Diploma in Drug Regulatory Affairs, Pfizer's "Molecule to Market" programme, Good Clinical Practice (GCP) certification, a data integrity/ALCOA+ course, and certifications in AI and data science applications in pharma.

It is not an official regulatory submission, not affiliated with or endorsed by any regulatory authority or pharmaceutical company, and not a real Periodic Safety Update Report (PSUR). Where document excerpts are written in a regulatory or PV style (Section 3 and Section 4), they are original writing produced for this project to demonstrate familiarity with that format — not text copied from or representing any real company's confidential records.

Every factual claim about domperidone's regulatory history and safety data in this report is drawn from public sources: European Medicines Agency (EMA) referral documentation, published peer-reviewed pharmacovigilance analysis, and public regulatory advisories. Full sourcing is listed in Section 5. All figures were current as of the cited publication dates — current regulatory status should be independently re-verified before this project is cited in any professional context.

A note on limitations, stated upfront rather than left for a reader to find: this project uses published aggregate safety data rather than a raw regulatory database extract, and Section 4's data model is a transparent, rule-based scoring illustration rather than a trained machine-learning model. Both choices, and the reasoning behind them, are explained in their respective sections.

Executive Summary

This report traces domperidone — a widely used anti-nausea medicine — through its complete regulatory lifecycle: authorization, real-world safety surveillance, signal detection, and regulatory action. It is built entirely from public sources: EMA referral documentation, published FAERS-based pharmacovigilance analysis, and regulatory records from the EU, UK, and Philippines.

The goal is to demonstrate integrated competency across three functions that a CRO, Big 4 life-sciences advisory practice, or pharmaceutical MNC expects an entry-level associate to move between: Regulatory Affairs, Pharmacovigilance, and Data Analysis.

Section 1 — Regulatory Landscape & Gap Analysis

1.1 Background

Domperidone is a dopamine antagonist used for nausea, vomiting, and — historically — dyspepsia and reflux symptoms. It has been marketed in the EU and other regions for decades, but was never approved by the US FDA for any indication, itself a useful regulatory comparison point.

1.2 Timeline of Regulatory Action

Date	Event
1985	Injectable form of domperidone withdrawn due to cardiac side effects
Pre-2014	Product information updated multiple times with cardiac warnings following case reports
2014 (request by Belgium)	Belgian medicines authority requests EMA review over continued cardiac safety concerns
March 3–6, 2014	EMA's PRAC completes an in-depth benefit-risk review
April 2014	CMDh endorses PRAC's recommendations
September 1, 2014	EMA final decision: restricted to nausea/vomiting relief only; dose capped; duration limited
December 2019	UK MHRA restricts pediatric use following a dedicated safety study

1.3 Cross-Agency Comparison

Parameter	EU (EMA/PRAC)	UK (MHRA)	US (FDA)	India (CDSCO)
Approved indication	Nausea/vomiting only (post-2014)	Aligned with EU + pediatric restriction (2019)	Not approved (IND only, gastroparesis)	Single-ingredient domperidone remains approved for nausea/vomiting; several domperidone-contai

Parameter	EU (EMA/PRAC)	UK (MHRA)	US (FDA)	India (CDSCO)
				ning fixed-dose combinations (e.g., Paracetamol+Domperidone+Caffeine) have been banned by CDSCO on safety/efficacy grounds
Max daily dose	30 mg/day	Same as EU	N/A	30 mg/day (per published Indian prescribing references) — aligned with the EU cap
Key contraindications added	QT-prolongation history, cardiac disease, hepatic impairment, CYP3A4 inhibitors	Same as EU	N/A	Not independently verified on the official CDSCO label — flagged for further verification
Regulatory mechanism	Article 31 referral (safety-driven)	Follows EU + independent MHRA action	N/A — never authorized	FDC-specific bans via CDSCO gazette notification, rather than a single-drug label restriction

India's approach is a genuinely interesting regulatory contrast: rather than restricting the single-ingredient drug's dose and indication directly (as the EU did), CDSCO banned specific fixed-dose combination products containing domperidone, while the plain single-ingredient drug remains available at a dose cap aligned with the EU standard. The cardiac-specific contraindication language has not been independently verified against the official current CDSCO label and is flagged as such rather than assumed.

1.4 CTD-Style Summary (Module 2.5 style extract)

“Post-marketing surveillance identified a signal of QT-interval prolongation and ventricular arrhythmia associated with domperidone exposure, disproportionately affecting patients aged over 60 years, those on doses above 30 mg/day, and those on concomitant CYP3A4 inhibitors or other QT-prolonging agents. Benefit-risk re-assessment concluded that the benefit-risk balance remained positive only for short-term symptomatic relief of nausea and vomiting at reduced dose, prompting label restriction under Article 31 of Directive 2001/83/EC.”

Original summary written for this project; not quoted from any single source document.

Section 2 — Pharmacovigilance Signal Analysis

Data source: a published analysis of FDA Adverse Event Reporting System (FAERS) records for domperidone, January 2008–June 2014 (Ogbru et al., PubMed ID 27182647). Where raw case-level FAERS extraction wasn't accessible for this project, published aggregate FAERS figures are used and cited — the same practice a junior PV associate follows when a live database pull isn't available.

2.1 Reported Case Counts

Metric	Count	% of relevant cohort
Total nonfatal cardiac events reported	221	—
Nonfatal cardiac events with concomitant QT-interacting drug	162	73.3%
Reported deaths with concomitant QT-interacting drug	53 of 151	35.1%
Reported cardiac arrests with concomitant QT-interacting drug	16 of 61	26.2%

A smaller national dataset (Philippines FDA, 2024) reported 97 total suspected domperidone adverse reaction cases, of which 4 involved QT prolongation/irregular heartbeat — included to show how signal strength can look different in a smaller national reporting system versus a large aggregator like FAERS.

2.2 Signal Detection Logic Applied

- Numerator concern: 73.3% of nonfatal cardiac events involved a co-prescribed QT-interacting drug — far higher than expected if the association were random.
- Consistency across severity bands: the pattern holds across nonfatal events, deaths, and cardiac arrests, which strengthens signal credibility.
- Biological plausibility: domperidone's known QT-prolonging mechanism (hERG channel blockade) gives a plausible causal pathway, which EMA's PRAC cited directly.

2.3 Mock PSUR-Style Excerpt (original text, illustrative)

Section 3: Cumulative Safety Review — Cardiac Events

“During the review period, cardiac events associated with domperidone use, particularly QT-interval prolongation and ventricular arrhythmia, continued to be disproportionately associated with concomitant use of QT-interacting medicinal products and CYP3A4 inhibitors. The consistency of this association across nonfatal, fatal, and cardiac-arrest case subsets supports a causal contribution of drug-drug interaction to the observed cardiac risk. No new safety concerns were identified beyond those already reflected in current product information; however, the cumulative case pattern supports continued restriction of dose, treatment duration, and contraindication in patients receiving other QT-prolonging agents.”

2.4 Data Integrity Note (ALCOA+ Application)

- Attributable — every figure traced to a named, cited publication
- Original — the published FAERS aggregate is treated as the original source; no recalculated numbers presented as raw data
- Accurate — percentages checked against the source publication's stated denominators

- Contemporaneous — the analysis explicitly states its data window (2008–2014) rather than implying current status

Section 3 — Data & AI Analysis

3.1 Visualizing the Signal

Outcome	With QT-interacting drug	Without QT-interacting drug	Total	% with QT-interacting drug
Nonfatal cardiac events	162	59	221	73.3%
Deaths	53	98	151	35.1%
Cardiac arrests	16	45	61	26.2%

A clustered column chart of this table (built in Excel or shown in the companion project files) makes the association visually clear before a single statistic is quoted.

3.2 A Note on Rigor

The “% with QT-interacting drug” column looks like a clean story, but this is not a controlled comparison — the background co-prescription rate among domperidone users generally isn’t known from this table, so a true relative risk can’t be computed from it alone. Naming that limitation directly, rather than letting a reader assume more rigor than exists, is deliberate.

3.3 Risk-Scoring Feature Logic (illustrative, not a trained model)

Patient-level FAERS records aren’t accessible in this project, so a trained classifier isn’t honestly buildable here. Instead, this section demonstrates the feature logic a real “serious AE risk flagging” tool would use, based on the actual risk factors EMA and FDA named in the domperidone reviews:

Feature	Source of risk factor
Age > 60	Named explicitly in EMA PRAC review
Daily dose > 30 mg	EMA’s revised dose cap threshold
Concomitant QT-interacting drug (Y/N)	Central factor in FAERS analysis (Section 2)
Concomitant CYP3A4 inhibitor (Y/N)	Named in EMA contraindication list
Pre-existing cardiac history (Y/N)	Named in EMA contraindication list

A simple rule-based scoring system (1 point per risk factor present, flagged as “high monitoring priority” at 2 or more points) mirrors how real pharmacovigilance triage logic is often built before a model is trained on labeled outcome data — a defensible, honest use of AI/data-science coursework rather than an overstated claim of a trained ML model.

Section 4 — Conclusion: Why This Project Matters

A CRO, Big 4 life-sciences advisory practice, or pharma MNC hiring for an entry-level Regulatory, PV, or data-adjacent role is not testing for years of industry experience — it is testing whether a candidate can read and correctly interpret real regulatory documents, understand how safety signals are actually detected and escalated, handle data honestly including its limitations, and connect all of this into one coherent professional narrative.

This project applies certifications already completed — the Drug Regulatory Affairs diploma, Pfizer’s Molecule to Market programme, GCP, ALCOA/data integrity training, and AI/data-science coursework in pharma — to one real, traceable case, rather than leaving them as a list of course completions on a resume.

Section 5 — Sources

- European Medicines Agency, “Domperidone Article 31 referral” (EMA/465179/2014) and EMA domperidone referral page
- European Medicines Agency, PRAC meeting highlights, 3–6 March 2014
- Ogbru et al., “Domperidone is Commonly Prescribed With QT-Interacting Drugs,” PubMed ID 27182647
- Philippines FDA Advisory No. 2024-0196
- UK MHRA pediatric domperidone restriction, December 2019
- CIMS/MIMS India, Domperidone dosing reference
- List of Fixed Dose Combinations banned by CDSCO (gazette notifications), via drugscontrol.org compilation